

Program : Diploma in Renewable Energy	
Course Code: 5293B	Course Title: Electric Power Generation and Transmission
Semester: 5	Credits: 4
Course Category : Program Elective Course	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To explain the various methods of power generation.
- To identify the basic knowledge of the economic considerations in generation of the electric power.
- To understand the electrical and mechanical characteristics of the electric power transmission.
- To identify the fundamental concepts of electric power distribution system and understand the use of underground cables and insulators in the power distribution system.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Fundamental knowledge in the electrical technology and renewable energy		Basic Electrical Technology and renewable sources.	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Illustrate the various methods of power generation.	15	Applying
CO2	Apply the economic considerations in generation of electric power.	14	Applying
CO3	Explain the basic concepts of electrical and mechanical characteristics of electrical power transmission	14	Understanding

CO4	Summarize the concepts of electric power distribution and outline the use of underground cables and insulators used in the power distribution system.	15	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the various methods of power generation.		
M1.01	Explain the generation of electrical energy	2	Understanding
M1.02	Outline of the different power stations	8	Understanding
M1.03	Compare various power plants	1	Understanding
M1.04	Solve the total electrical energy generated in the various power plants	4	Applying
Contents: - Generation of electrical energy-sources of energy. Generating stations –Steam power station-Schematic arrangement, choice of site, efficiency, equipments, advantages and disadvantages. Hydroelectric power station-Schematic arrangement, choice of site, constituents, advantages and disadvantages. Diesel power station-Schematic arrangement, advantages and disadvantages.			

Nuclear power station-Schematic arrangement, selection of site, advantages and disadvantages. Comparison of various power plants. Calculation of total electrical energy generated in various power plants - problems.			
CO2	Identify the economic considerations in generation of electric power.		
M2.01	Organize the basic concepts of variable load on power system.	5	Applying
M2.02	Interpret the fundamental concepts of base load and peak load on the power station	2	Understanding
M2.03	Identify the economics of generation of electric power	3	Applying
M2.04	Make use of different types of tariff to calculate electricity charges	4	Applying
	Series Test – I	1	
Contents : Variable load on power station - load curve –important terms and factors –simple problems. Base load and peak load on power station-Interconnected grid system. Economics of power generation-cost of electrical energy-importance of high load factor-simple problems- Tariffs – objectives of tariffs– types of tariff- simple problems.			
CO3	Explain the basic concepts of electrical and mechanical characteristics of the electric power transmission.		
M3.01	Describe the electric supply system.	4	Understanding
M3.02	Summarize the importance of overhead lines in the power transmission	4	Understanding
M3.03	Identify corona effect and sag in overhead line	3	Understanding
M3.04	Illustrate electrical characteristics of the transmission line.	4	Understanding
Contents :- Electric supply system – Typical A.C power supply scheme – comparison of D.C and A.C transmission-Advantages of high transmission voltage-Various systems of power transmission.			

Overhead lines-main components-conductor materials-line supports-insulators			
Corona effect and Sag in overhead lines-simple problems			
Classification of overhead transmission lines - Line losses - Voltage regulation - Transmission efficiency -Ferranti effect – Skin effect.			
CO4	Summarize the concepts of electric power distribution and outline the use of underground cables and insulators in the power distribution system.		
M4.01	Describe the basic concepts of the electric power distribution system	7	Understanding
M4.02	Explain the fundamental concepts of Underground cables	4	Understanding
M4.03	Compare overhead and underground transmission systems	1	Understanding
M4.04	Summarize the use of different insulators	3	Understanding
	Series Test – II	1	
Contents : - Distribution system-classification of distribution systems-AC distribution-D.C distribution-Methods of obtaining 3 wire D.C distribution system-Connection schemes of A.C distribution system-requirements of a distribution system Underground Cables-Construction-insulating materials for cables-classification of cables-cables for three phase service. Compare overhead and underground transmission systems. Insulators-Types of insulators-potential distribution over suspension insulator string-string efficiency.			

Text / Reference

T/R	Book Title/Author
T1	VK Mehtha. RohitMehtha. Principles of Power System: S Chand Publishing
T2	C.L Wadhwa ‘Electrical Power Systems’,New Age International publishers
R1	J B Gupta. A course in Power Systems: S K Kataria&sons.

Online Resources

Sl.No	Website Link
1	https://www.digimat.in/nptel/courses/video/108102047/L01.html
2	https://www.electricaltechnology.org/category/power-system.
3	https://www.brainkart.com/article/Generation,-Transmission-and-Distribution-of-Electric-Power_12341/